

R Code

```
1  install.packages("tidyverse")
2  install.packages("readxl")
3  install.packages("lubridate")
4  install.packages("janitor")
5
6  library(tidyverse)
7  library(readxl)
8  library(lubridate)
9  library(janitor)
10
11  customer_data <- read_csv("C:/Users/hites/OneDrive/Desktop/Hitesh/Portfolio/Additional
Skills/Data Analytics Job Simulation - Quantum/data/QVI_purchase_behaviour.csv")
12  view(customer_data)
13  transaction_data <-
  read_excel("C:/Users/hites/OneDrive/Desktop/Hitesh/Portfolio/Additional Skills/Data
Analytics Job Simulation - Quantum/data/QVI_transaction_data.xlsx")
14  View(transaction_data)
15  glimpse(transaction_data)
16  glimpse(customer_data)
17
18  head(transaction_data)
19  head(customer_data)
20
21  transaction_data <- clean_names(transaction_data)
22  customer_data <- clean_names(customer_data)
23
24  glimpse(transaction_data)
25  glimpse(customer_data)
26
27  colSums(is.na(transaction_data))
28  colSums(is.na(customer_data))
29
30  sum(duplicated(transaction_data))
31  sum(duplicated(customer_data))
32
33  transaction_data[duplicated(transaction_data), ]
34
35  transaction_data <- transaction_data %>%
36    distinct()
37
38  sum(duplicated(transaction_data))
39  # Data Quality Findings
40  # One fully duplicated transaction record was identified.
41  # The duplicate was removed to avoid overstating sales,
42  # transaction counts and customer purchasing metrics.
43
44  colSums(is.na(transaction_data))
45  colSums(is.na(customer_data))
46  # Missing Value Check
47  # No missing values were identified in either the transaction
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48 # dataset or customer dataset.
49 # No imputation or data removal was required.
50
51 summary(transaction_data$date)
52 head(transaction_data$date)
53
54 transaction_data %>%
55   count(prod_qty) %>%
56   arrange(desc(prod_qty))
57
58 transaction_data %>%
59   count(prod_name, sort = TRUE)
60
61 transaction_data <- transaction_data %>%
62   mutate(date = as.Date(date, origin = "1899-12-30"))
63
64 head(transaction_data$date)
65 summary(transaction_data$date)
66 # Date Format Check
67 # Transaction dates were stored as Excel serial values.
68 # Dates were converted to standard Date format for analysis.
69
70 transaction_data %>%
71   filter(prod_qty == 200)
72 # Outlier Investigation
73 # Two transactions were identified with a purchase quantity of 200 packs.
74 # Both transactions belonged to the same loyalty customer and represented
75 # purchases substantially larger than the rest of the dataset.
76 # As the objective is to understand typical customer purchasing behaviour,
77 # these transactions were considered outliers and removed from analysis.
78 transaction_data <- transaction_data %>%
79   filter(prod_qty < 200)
80 max(transaction_data$prod_qty)
81
82 transaction_data %>%
83   filter(str_detect(str_to_lower(prod_name), "salsa"))
84
85 nrow(
86   transaction_data %>%
87     filter(str_detect(str_to_lower(prod_name), "salsa"))
88 )
89
90 transaction_data %>%
91   filter(str_detect(str_to_lower(prod_name), "salsa")) %>%
92   count(prod_name, sort = TRUE)
93
94 transaction_data <- transaction_data %>%
95   filter(!str_detect(str_to_lower(prod_name), "salsa"))
96 nrow(transaction_data)
97 # Category Validation
98 # 18,094 transactions were identified as salsa products.
99 # As the analysis focuses specifically on the chip category,
100 # salsa products were excluded from further analysis.
101

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102 transaction_data <- transaction_data %>%
103   mutate(
104     pack_size = str_extract(prod_name, "\\d+[gG]"),
105     pack_size = str_remove(pack_size, "[gG]"),
106     pack_size = as.numeric(pack_size)
107   )
108
109 summary(transaction_data$pack_size)
110
111 transaction_data %>%
112   count(pack_size, sort = TRUE)
113
114 transaction_data <- transaction_data %>%
115   mutate(
116     brand = word(prod_name, 1)
117   )
118
119 transaction_data %>%
120   count(brand, sort = TRUE)
121
122 summary(transaction_data$pack_size)
123
124 transaction_data %>%
125   count(pack_size, sort = TRUE)
126
127 transaction_data %>%
128   count(brand, sort = TRUE)
129
130
131 transaction_data %>%
132   filter(is.na(pack_size)) %>%
133   distinct(prod_name)
134
135 sum(is.na(transaction_data$pack_size))
136 summary(transaction_data$pack_size)
137 transaction_data %>%
138   count(pack_size, sort = TRUE)
139 # Feature Engineering Check
140 # Initial pack size extraction produced 6,064 missing values.
141 # Investigation showed several product descriptions used an uppercase "G"
142 # rather than lowercase "g".
143 # The extraction logic was updated to capture both formats,
144 # eliminating all missing pack size values.
145
146 transaction_data %>%
147   count(brand, sort = TRUE) %>%
148   print(n = 50)
149
150 transaction_data %>%
151   filter(brand %in% c(
152     "Red", "RRD",
153     "Dorito", "Doritos",
154     "Smith", "Smiths",
155     "Grain", "GrnWves",

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156     "Infzns", "Infuzions",
157     "Snbts", "Sunbites",
158     "NCC", "Natural",
159     "WW", "Woolworths"
160   )) %>%
161   distinct(prod_name) %>%
162   arrange(prod_name)
163
164   transaction_data <- transaction_data %>%
165     mutate(
166       brand = case_when(
167         brand == "Red" ~ "RRD",
168         brand == "Dorito" ~ "Doritos",
169         brand == "Smith" ~ "Smiths",
170         brand == "Grain" ~ "Grain Waves",
171         brand == "GrnWves" ~ "Grain Waves",
172         brand == "Infzns" ~ "Infuzions",
173         brand == "Snbts" ~ "Sunbites",
174         brand == "Natural" ~ "Natural Chip Co",
175         brand == "NCC" ~ "Natural Chip Co",
176         brand == "Woolworths" ~ "WW",
177         TRUE ~ brand
178       )
179     )
180
181   transaction_data %>%
182     count(brand, sort = TRUE)
183   # Brand Standardisation
184   # Product descriptions were used to derive brand names.
185   # Several inconsistencies were identified (e.g. Dorito vs Doritos,
186   # Smith vs Smiths, Infzns vs Infuzions).
187   # Brand names were standardised to ensure accurate aggregation and
188   # analysis of brand performance.
189
190
191
192   combined_data <- transaction_data %>%
193     left_join(customer_data, by = "lylty_card_nbr")
194
195   colSums(is.na(combined_data))
196
197   combined_data %>%
198     summarise(
199       total_sales = sum(tot_sales),
200       total_transactions = n(),
201       total_customers = n_distinct(lylty_card_nbr),
202       total_quantity = sum(prod_qty),
203       avg_transaction_value = sum(tot_sales) / n(),
204       avg_units_per_transaction = sum(prod_qty) / n()
205     )
206
207
208   sales_by_segment <- combined_data %>%
209     group_by(lifestage, premium_customer) %>%

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210     summarise(
211       total_sales = sum(tot_sales),
212       customers = n_distinct(lylty_card_nbr),
213       transactions = n(),
214       total_qty = sum(prod_qty),
215       avg_sales_per_customer = total_sales / customers,
216       avg_units_per_customer = total_qty / customers,
217       avg_price_per_unit = total_sales / total_qty
218     ) %>%
219     arrange(desc(total_sales))
220
221 sales_by_segment
222
223
224 colSums(is.na(combined_data))
225
226 combined_data %>%
227   summarise(
228     total_sales = sum(tot_sales),
229     total_transactions = n(),
230     total_customers = n_distinct(lylty_card_nbr),
231     total_quantity = sum(prod_qty)
232   )
233
234 sales_by_segment
235
236
237
238 sales_summary <- combined_data %>%
239   group_by(lifestage, premium_customer) %>%
240   summarise(
241     total_sales = sum(tot_sales),
242     .groups = "drop"
243   )
244
245 #Chart 1 – Total Sales
246
247 ggplot(
248   sales_summary,
249   aes(
250     x = lifestage,
251     y = total_sales,
252     fill = premium_customer
253   )
254 ) +
255   geom_col(position = "dodge") +
256   labs(
257     title = "Total Sales by Customer Segment",
258     x = "Lifestage",
259     y = "Total Sales"
260   ) +
261   theme_bw() +
262   theme(axis.text.x = element_text(angle = 45, hjust = 1))
263

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264 # Chart - 2 Number of Customers
265
266 customer_summary <- combined_data %>%
267   group_by(lifestage, premium_customer) %>%
268   summarise(
269     customers = n_distinct(lylty_card_nbr),
270     .groups = "drop"
271   )
272
273 ggplot(
274   customer_summary,
275   aes(
276     x = lifestage,
277     y = customers,
278     fill = premium_customer
279   )
280 ) +
281   geom_col(position = "dodge") +
282   labs(
283     title = "Number of Customers by Segment",
284     x = "Lifestage",
285     y = "Customers"
286   ) +
287   theme_bw() +
288   theme(axis.text.x = element_text(angle = 45, hjust = 1))
289
290 # Chart 3 – Average Units per Customer
291
292 units_summary <- combined_data %>%
293   group_by(lifestage, premium_customer) %>%
294   summarise(
295     customers = n_distinct(lylty_card_nbr),
296     total_units = sum(prod_qty),
297     avg_units_per_customer = total_units / customers,
298     .groups = "drop"
299   )
300
301 ggplot(
302   units_summary,
303   aes(
304     x = lifestage,
305     y = avg_units_per_customer,
306     fill = premium_customer
307   )
308 ) +
309   geom_col(position = "dodge") +
310   labs(
311     title = "Average Units per Customer",
312     x = "Lifestage",
313     y = "Units per Customer"
314   ) +
315   theme_bw() +
316   theme(axis.text.x = element_text(angle = 45, hjust = 1))
317

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318 # Chart 4 – Average Price per Unit
319
320 price_summary <- combined_data %>%
321   group_by(lifestage, premium_customer) %>%
322   summarise(
323     avg_price_per_unit = sum(tot_sales) / sum(prod_qty),
324     .groups = "drop"
325   )
326
327 ggplot(
328   price_summary,
329   aes(
330     x = lifestage,
331     y = avg_price_per_unit,
332     fill = premium_customer
333   )
334 ) +
335   geom_col(position = "dodge") +
336   labs(
337     title = "Average Price per Unit by Segment",
338     x = "Lifestage",
339     y = "Average Price"
340   ) +
341   theme_bw() +
342   theme(axis.text.x = element_text(angle = 45, hjust = 1))
343
344
345 combined_data <- combined_data %>%
346   mutate(unit_price = tot_sales / prod_qty)
347
348 mainstream <- combined_data %>%
349   filter(
350     premium_customer == "Mainstream",
351     lifestage %in% c(
352       "YOUNG SINGLES/COUPLES",
353       "MIDAGE SINGLES/COUPLES"
354     )
355   ) %>%
356   pull(unit_price)
357
358 other <- combined_data %>%
359   filter(
360     premium_customer %in% c("Budget", "Premium"),
361     lifestage %in% c(
362       "YOUNG SINGLES/COUPLES",
363       "MIDAGE SINGLES/COUPLES"
364     )
365   ) %>%
366   pull(unit_price)
367
368 t.test(mainstream, other)
369
370 # Creating Target Segment
371

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```

372 target_segment <- combined_data %>%
373   filter(
374     lifestage == "YOUNG SINGLES/COUPLES",
375     premium_customer == "Mainstream"
376   )
377
378 # Brand Preference of Target Segment
379
380 target_brand <- target_segment %>%
381   count(brand) %>%
382   mutate(
383     target_prop = n / sum(n)
384   )
385
386 other_brand <- combined_data %>%
387   filter(
388     !(lifestage == "YOUNG SINGLES/COUPLES" &
389       premium_customer == "Mainstream")
390   ) %>%
391   count(brand) %>%
392   mutate(
393     other_prop = n / sum(n)
394   )
395
396 brand_affinity <- target_brand %>%
397   left_join(other_brand, by = "brand") %>%
398   mutate(
399     affinity = target_prop / other_prop
400   ) %>%
401   arrange(desc(affinity))
402
403 brand_affinity
404
405 # Pack Size Affinity
406
407 target_pack <- target_segment %>%
408   count(pack_size) %>%
409   mutate(
410     target_prop = n / sum(n)
411   )
412
413 # Other Population Pack Preference
414 other_pack <- combined_data %>%
415   filter(
416     !(lifestage == "YOUNG SINGLES/COUPLES" &
417       premium_customer == "Mainstream")
418   ) %>%
419   count(pack_size) %>%
420   mutate(
421     other_prop = n / sum(n)
422   )
423
424 # Compare
425 pack_affinity <- target_pack %>%

```

```

426     left_join(other_pack, by = "pack_size") %>%
427     mutate(
428       affinity = target_prop / other_prop
429     ) %>%
430     arrange(desc(affinity))
431
432     pack_affinity
433
434
435     head(brand_affinity, 10)
436     # Insights:
437     # Mainstream Young Singles/Couples show a stronger preference for
438     # Tyrrells, Twisties, Doritos, Tostitos, Kettle and Pringles
439     # compared to the rest of the customer base.
440     #
441     # These brands over-index within the target segment and represent
442     # potential opportunities for targeted promotions and merchandising.
443     head(pack_affinity, 10)
444     # Insights:
445     # Mainstream Young Singles/Couples demonstrate a stronger preference
446     # for larger pack sizes, particularly 270g, 330g and 380g packs.
447     #
448     # This suggests that larger sharing packs are more appealing to
449     # this customer segment and may present opportunities for
450     # category growth.
451
452
453
454     # Recommendation:
455     # Mainstream Young Singles/Couples should be considered the primary
456     # target segment for the chips category. This segment contributes
457     # significant sales, pays a higher average price per unit, and
458     # shows stronger preferences for premium brands and larger pack sizes.
459     #
460     # Category growth opportunities may include increasing visibility
461     # of preferred brands such as Doritos, Kettle and Pringles, while
462     # promoting larger pack sizes that align with this segment's
463     # purchasing behaviour.

```